

Introdução às Redes de Comunicação

Teste sobre Winsock TCP e Multiplexagem de E/S - Ficha de consulta

CRIAÇÃO, ASSOCIAÇÃO A UM PORTO LOCAL E FECHO DE SOCKETS WINDOWS

```
SOCKET socket(int af, int type, int protocol); /* PF_INET, SOCK_STREAM, IPPROTO_TCP */  
  
int bind(SOCKET s, const struct sockaddr *name, int namelen);  
  
int closesocket(SOCKET s);
```

ESTABELECIMENTO DE LIGAÇÕES TCP

```
int connect(SOCKET s, const struct sockaddr *name, int namelen);  
  
SOCKET accept(SOCKET s, struct sockaddr *from, int *fromlen);
```

INDICAÇÃO DE ERRO E CÓDIGOS DE ERRO

```
SOCKET_ERROR  
  
INVALID_SOCKET  
  
int WSAGetLastError(void); /* WSAETIMEDOUT */
```

LOCALIZAÇÃO E CONVERSÃO DE FORMATOS

```
struct sockaddr_in a; /* a.sin_family, a.sin_addr.s_addr, a.sin_port */  
  
u_short htons(u_short hostshort); /* host to network short */  
  
u_long htonl(u_long hostlong); /* host to network long */  
  
u_short ntohs(u_short netshort); /* network to host short */  
  
u_long ntohl(u_long netlong); /* network to host long */  
  
unsigned long inet_addr(const char *cp);  
  
char* inet_ntoa(struct in_addr in); /* network to ascii */
```

RESOLUÇÃO DE NOMES

INADDR_NONE

```
struct hostent* gethostbyname(char *name);  
  
/*  
  
** struct hostent info;  
  
** struct sockaddr_in addr;  
  
** ...  
  
** memcpy(&(addr.sin_addr.s_addr), info->h_addr, info->h_length);  
  
** ...  
  
*/
```

ENVIO E RECEPÇÃO DE DATAGRAMAS

```
int send(SOCKET s, const char *buf, int len, int flags);  
  
int recv(SOCKET s, char *buf, int len, int flags);
```

MULTIPLEXAGEM DE SOCKETS

```
int select(32, fd_set *readfds, NULL, NULL, struct timeval *timeout);  
  
FD_ZERO(&set);  
  
FD_SET(s, &set);  
  
FD_ISSET(s, &set);  
  
struct timeval { long tv_sec; long tv_usec;}
```

OBTENÇÃO DE INFORMAÇÃO LOCAL E REMOTA ASSOCIADA AOS SOCKETS

```
int getpeername(SOCKET s, struct sockaddr *name, int *namelen);  
  
int getsockname(SOCKET s, struct sockaddr *name, int *namelen);  
  
int strcmp(const char *s1, const char *s2);  
  
char * strcpy_s(char * strDestination, int sizeStrDestination, const char * strSource);
```

CONFIGURAÇÃO DE OPÇÕES/PARÂMETROS

```
int setsockopt(SOCKET s, int level, int optname, const char *optval, int optlen);  
  
/* level = SOL_SOCKET, optname = SO_RCVTIMEO, optval = (char *)&timeoutMsec (timeoutMsec do tipo  
DWORD) */
```

CRIAÇÃO DE THREADS

```
HANDLE WINAPI CreateThread(  
    _In_opt_ LPSECURITY_ATTRIBUTES lpThreadAttributes,  
    _In_     SIZE_T dwStackSize,  
    _In_     LPTHREAD_START_ROUTINE lpStartAddress,  
    _In_opt_ LPVOID lpParameter,  
    _In_     DWORD dwCreationFlags,  
    _Out_opt_ LPDWORD lpThreadId  
);  
  
/*  
** void AtendeCliente(LPVOID param);  
** SECURITY_ATTRIBUTES sa;  
** DWORD thread_id;  
** ...  
** sa.nLength=sizeof(sa);  
** sa.lpSecurityDescriptor=NULL;  
** ...  
** CreateThread(&sa, 0, (LPTHREAD_START_ROUTINE)AtendeCliente, (LPVOID)param,  
** (DWORD)0, &thread_id);  
*/  
  
void ExitTread(...);
```